

EXPERIMENT - 1

Date: 2082-05-04

TO DETERMINE THE VOLUME OF A GIVEN HOLLOW TUBE BY USING VERNIER CALLIPERS AND HENCE CALCULATE ITS DENSITY.

APPARATUS REQUIRED:

1. Vernier Callipers
2. Given tube
3. Physical Balance

THEORY:

The vernier constant (V.C) of a Vernier calipers is the value of the difference between a main scale division and a Vernier division expressed in terms of the value of 1 main scale divisions (S).

Hence, if n Vernier divisions coincide with $(n-1)$ main scale divisions then

we have, 1 Vernier division coincides with $\frac{(n-1)}{n}$ main scale divisions.

∴ Difference between a main scale division and 1 Vernier scale division.

$$= 1 - \frac{(n-1)}{n} = \frac{1}{n}$$

∴ Vernier Constant (V.C.) = $\frac{1}{n} \times \text{Value of 1 main scale division (S)}$

$$= \frac{S}{n}$$

Let, l = length of tube

d_1 = internal diameter of the tube

d_2 = external diameter of the tube

Then, internal volume of the tube = $\frac{\pi}{4} d_1^2 l = V_{\text{int}}$

External volume of the tube = $\frac{\pi}{4} d_2^2 l = V_{\text{ext}}$

∴ Volume of tube = $V_{\text{ext}} - V_{\text{int}}$

$$= \frac{\pi}{4} d_2^2 l - \frac{\pi}{4} d_1^2 l = \frac{\pi}{4} (d_2^2 - d_1^2) l$$

If m be the mass of the tube, then the density of the tube is

$$\text{Density } (\rho) = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{i.e. } \rho = \frac{m}{V}$$

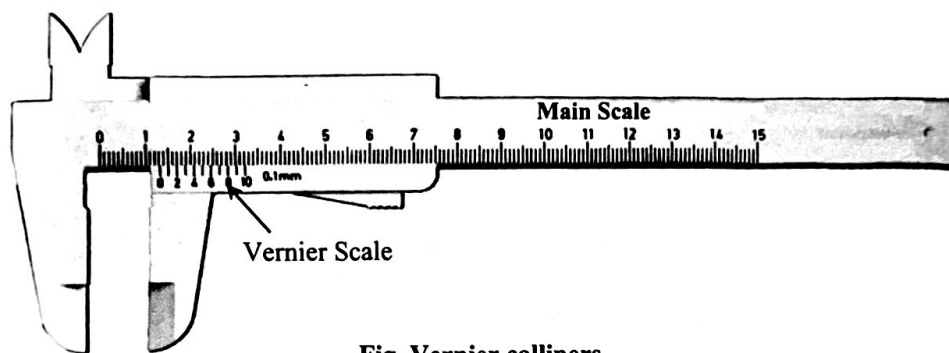


Fig. Vernier callipers

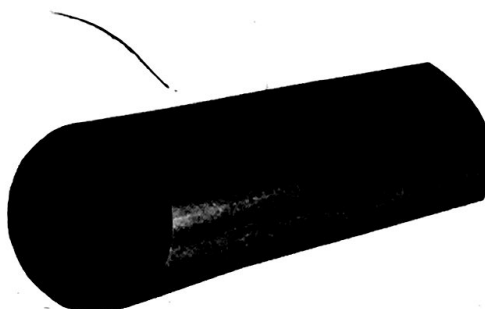


Fig. Hollow tube

OBSERVATIONS:

The zero of the Vernier scale Coincides with the zero of the main scale.

∴ Instrument Error = Zero

Zero Correction = Zero

The total number of division on Vernier scale (n) = 10

Value of 10 divisions of the main scale = 10 mm

Value of 1 divisions of the main scale (S) = 1 mm

10 Vernier scale division coincide with 9 main scale division

∴ 1 Vernier scale division coincide with 9/10 main scale division

∴ Vernier constant (V.C.) = (1 m.s.d. - 1 v.s.d.) × value of one main scale division (S.)

Now, Vernier constant (V.C.) = (1 - 0.9) mm

= 0.1 mm

= 0.01 cm

Mass of the given tube (m) = 4.40 g

No. of obs.	Obs. For	Reading of the			Total = (a+b) cm	Apparent mean (cm)	Corrected mean
		Main scale (a) cm	Vernier scale (y) divs.	Value of Vernier scale = y × V.C. (b) cm			
1	Length	5.1 cm	7	0.07	5.17	5.16	$l = 5.16$ cm
2		5.1 cm	9	0.09	5.19		
3		5.1 cm	6	0.06	5.16		
4		5.1 cm	4	0.04	5.14		
5		5.1 cm	6	0.06	5.16		
1	External diameter	1.8 cm	3	0.03	1.83	1.85	$d_2 = 1.85$ cm
2		1.8 cm	8	0.08	1.88		
3		1.8 cm	1	0.01	1.81		
4		1.8 cm	8	0.08	1.88		
5		1.8 cm	9	0.09	1.89		
1	Internal diameter	1.7 cm	2	0.02	1.72	1.73	$d_1 = 1.73$ cm
2		1.7 cm	1	0.01	1.71		
3		1.7 cm	2	0.02	1.72		
4		1.7 cm	3	0.03	1.73		
5		1.7 cm	9	0.09	1.79		

CALCULATION:

$$\begin{aligned}
 \therefore \text{Volume of tube (V)} &= \frac{\pi}{4}(d_2^2 - d_1^2)l \\
 &= \frac{22}{7} \times (3.45^2 - 3.006^2) \\
 &= 1.80 \text{ cm}^3
 \end{aligned}$$

$$\text{Now, Density } (\rho) = \frac{m}{V} = \frac{4.40}{1.80} = 2.44 \text{ gm/cm}^3$$

RESULT:

The volume and density of the material of the given hollow cylinder was found to be 1.80 cm^3 and 2.44 gm/cm^3 respectively

PERCENTAGE ERROR:

For Aluminium

Standard value for the density, ρ , (S.V.) = 2.7 g/cm^3

$$\begin{aligned} \% \text{ Error} &= \left| \frac{\text{S.V.} - \text{O.V.}}{\text{S.V.}} \right| \times 100\% \\ &= \frac{2.7 - 2.44}{2.7} \times 100\% \\ &= 9.78\% \end{aligned}$$

CONCLUSION:

The experiment successfully determined the volume of the hollow tube using vernier callipers.

SOURCES OF ERROR:

- Excessive pressure applied to Jaws of calipers.
- Impure form of material of Tube
- Carelessness in the experiment.
- The shape of tube may not be symmetrical

PRECAUTIONS:

1. The mass of tube should be taken carefully.
2. The Jaws should not be pressed too hard.
3. The zero error should be noticed carefully.
4. Use Magnified to find out the coinciding division.
5. Observations should be taken at different points on the tube

VIVA QUESTIONS:

1. What do you mean by least count of an instrument?
2. Why it is called Vernier calipers?
3. What is Vernier constant?
4. What do you mean by zero error in Vernier calipers?
5. How can you make Vernier calipers more precise?
6. Can you measure the depth of a cylinder with the help of Vernier calipers?
7. What are the maximum and minimum lengths that can be measured with the help of Vernier calipers?
8. A student records a length measurement made by a meter scale as 4.57 cm . Is he correct in his observation?

1. → The smallest measurement that can be accurately read with it is least count of an instrument.

2. → A french mathematician named pierre vernier invented it and it is named after him.

3. → The minimum length which can be measured accurately with the help of a given vernier calipers is called vernier constant.

4. → When the two Jaws of the calipers are brought into contact, the zero of the vernier scale should coincide with the zero of the main scale. If they do not coincide, it has a zero error. If the zero of the vernier scale is to the right or left of the zero of the main scale, the instrumental error is respectively positive or negative.

5. → We have, $V.C = \frac{\text{Value of 1 M.S.D}}{\text{No. of V.S.D}}$

The instrument can be made more sensitive i.e more accurate by decreasing V.C. This can be done by decreasing the value of a main scale division or by increasing the no. of divisions on vernier scale.

6. → Yes, we can measure the depth of a cylinder with the help of vernier caliper by inserting the depth rod into the hole we are measuring.

7. $\rightarrow$ Maximum length that can be measured with vernier caliper is 15 cm and minimum length ~~of~~ is 0.1 mm

8. $\rightarrow$ No, the student isn't correct in his observation. It is because least count for meter scale is 0.1 cm.

LC = 1 mm